Tutorial 3: FINA2205
Financial Regression Analysis 1

Zirui Song

zirui.song@uwa.edu.au

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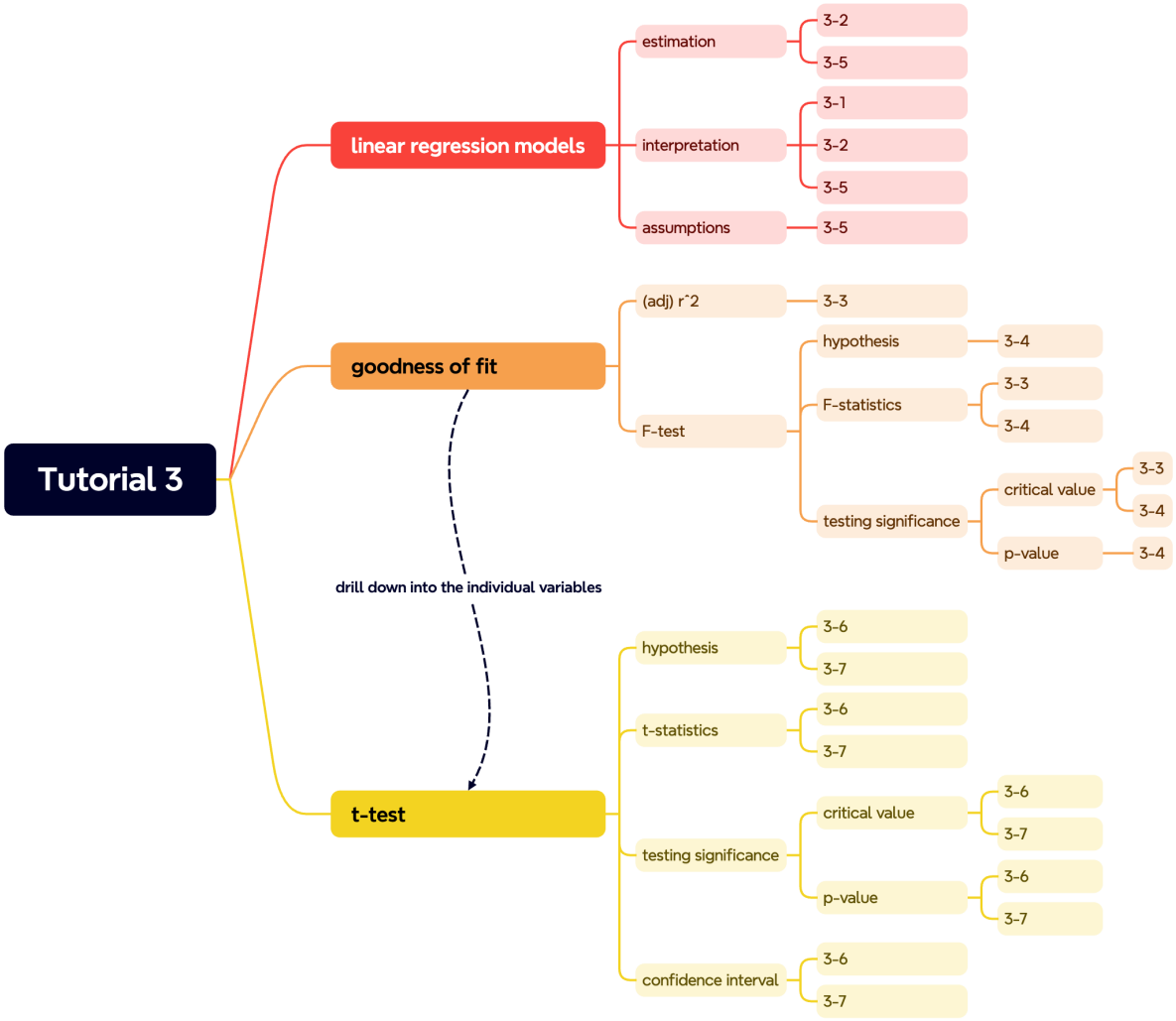
Agenda

- 7 problems



Recap: linear regression

- Last week, we used expected value to summarize what could happen for a single variable.
- We also use correlation to see if two variables move together.
- However, correlation only tells us they are related, but it doesn't give us a formula to predict Y from X.
- Question: Can we predict an **exact** Exam Score (Y) based on Study Hours (X)?
- No, because different students with the exact same study hours will still get different scores.

Recap: linear regression

- Since we cannot predict the exact outcome for an individual, we predict the expected value of Y given x.
- Regression estimates the line that best represents the **average** relationship in the sample data.
- Since we rarely know the true population relationship, we use a sample of observations, which typically contains noise and variation.
- Regression models come in different forms, including linear regression and extensions such as logistic regression.

Recap: linear regression

Simple linear regression

- Use **one** predictor to explain/predict an outcome
- Example: predict exam score from study hours
- Equation: $\hat{Y} = b_0 + b_1 X$
- Meaning of the parts:
  - $\hat{Y}$: predicted outcome
  - b_0: $intercept = predicted\ value\ of\ Y\ when\ X = 0$
  - b_1: $slope = expected\ change\ in\ Y\ when\ X\ increases\ by\ one\ unit$

Recap: linear regression

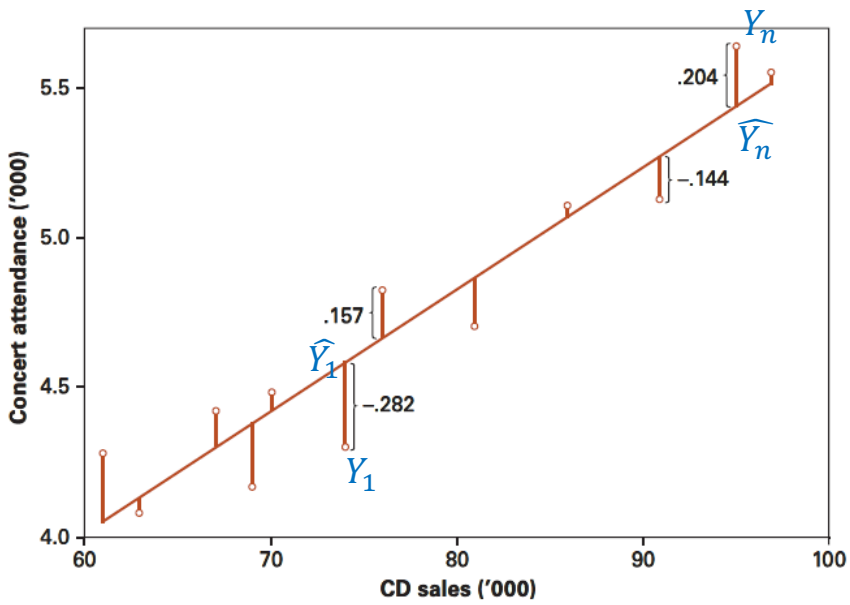
How do we choose the best-fitting line? (estimation)

- In simple linear regression, many different lines could be drawn through the data.
- We need a rule to choose the **best-fitting line**.
- **Ordinary Least Squares (OLS)** chooses the line that minimizes the **sum of squared residuals**.
- It gives us the line with the **smallest overall gap** between actual values and fitted values.

$$\text{Min (SSE)} = \sum (Y - \hat{Y})^2$$

Recap: linear regression

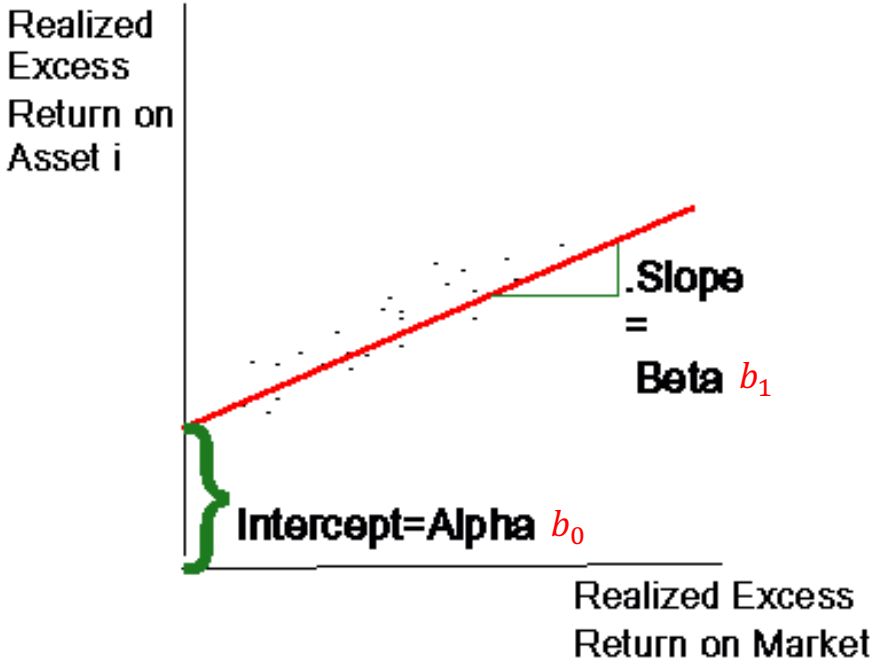
Estimation



$$\sum (Y - \hat{Y})^2 = (-.282)^2 + (.157)^2 + (-.144)^2 + (.204)^2 + \cdots$$

$$(Y_1 - Y_1)^2 + (Y_2 - Y_2)^2 + \cdots$$

Interpretation

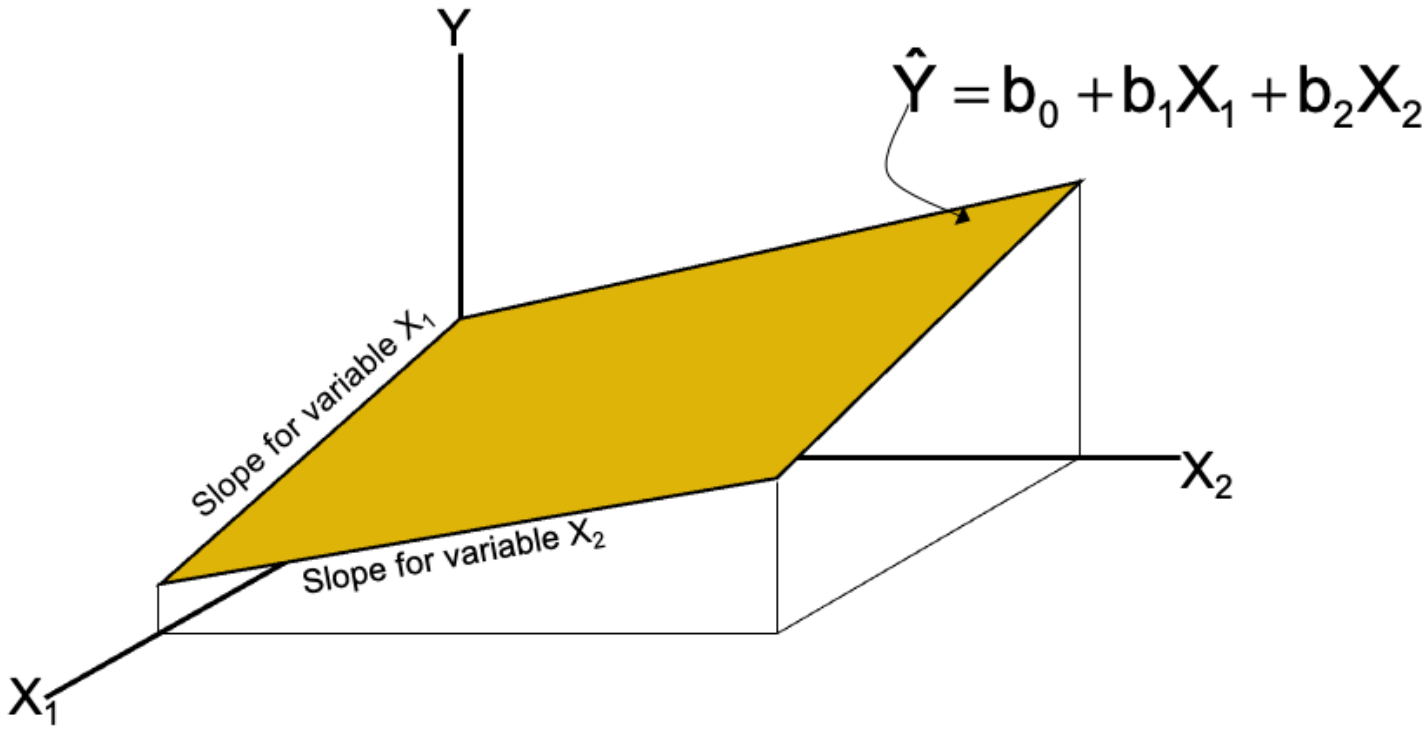


Recap: linear regression

Multiple regression

- Other factors may matter too, such as sleep hours, attendance, or prior knowledge.
- Use two or more predictors to explain/predict the same outcome
- Example: predict exam score from study hours and sleep hours
- Equation: $\hat{Y} = b_0 + b_1 X_1 + b_2 X_2$
- The intercept estimates the value of Y when **both** X_1 and X_2 are 0.
- Each slope tells us the effect of one predictor, **holding the other predictor(s) constant.**

Recap: linear regression



3-1

1. Use the following multiple regression equation.

$$\hat{Y}_i = 40 - 7X_{1i} + 2X_{2i}$$

a. Interpret the meaning of the slopes.
b. Interpret the meaning of the Y intercept.

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a. Interpret the meaning of the slopes.

For each one-unit increase in X_1, Y will (1) _____ an estimated _____ units, holding (2) _____ constant. For each one-unit increase in X_2, Y will (3) _____ an estimated _____ units, holding (4) _____ constant.

(1) ○ decrease
○ increase

(2) ○ X_1
○ X_2
○ Y

(3) ○ decrease
○ increase

(4) ○ X_1
○ Y
○ X_2

3-1 solution

a.

For each one-unit increase in X_1, Y will <u>decrease</u> an estimated <u>7</u> units, holding <u>X_2</u> constant.

For each one-unit increase in X_2, Y will <u>increase</u> an estimated <u>2</u> units, holding <u>X_1</u> constant.

3-1 (cont.)

$\hat{Y}_i = 40 - 7X_{1i} + 2X_{2i}$

b. Interpret the meaning of the Y intercept. Choose the correct answer below.

- **A.** The Y-intercept 40 estimates the value of Y when X_1 is 0.
- **B.** The Y-intercept 40 estimates the value of Y when both X_1 and X_2 are 0.
- **C.** The Y-intercept 40 estimates the value of Y when X_2 is 0.
- **D.** The Y-intercept 7 estimates the value of Y when both X_1 and X_2 are 0.
- **E.** The Y-intercept 2 estimates the value of Y when both X_1 and X_2 are 0.
- **F.** The Y-intercept 7 estimates the value of Y when X_1 is 0.
- **G.** The Y-intercept 2 estimates the value of Y when X_2 is 0.

3-1 solution (cont.)

b.

- B. The Y-intercept 40 estimates the value of Y when both X_1 and X_2 are 0.

3-2

2. A certain town is located approximately 25 miles east of a large city. The data organized below include the appraised value (in thousands of dollars), land area of the property in acres, and age, in years, for a sample of 20 single-family homes located in the town. Develop a multiple linear regression model to predict appraised value based on land area of the property and age, in years. Complete parts (a) through (f).
[1] Click the icon to view the data table.

a. State the multiple regression equation. Let X_{1i} represent the land area of the property in acres and let X_{2i} age, in years.

$\hat{Y}_i =$ _____ + (_____)X_{1i} + (_____)X_{2i}

(Round to four decimal places as needed.)

b. Interpret the meaning of the slopes, b_1 and b_2, in this problem. Choose the correct answer below.

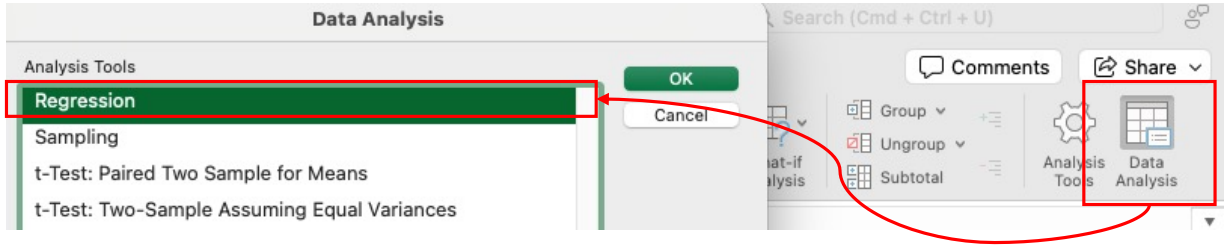
- ○ **A.** For a given age, each increase of 1 acre in land area is estimated to result in an increase in appraised value by $1000b_1$ dollars. For a given land area, each increase in one year in age is estimated to result in a decrease in appraised value by $1000b_2$ dollars.
- ○ **B.** For a given age, each increase of 1 acre in land area is estimated to result in an increase in appraised value by b_2 dollars. For a given land area, each increase in one year in age is estimated to result in a decrease in appraised value by b_1 dollars.
- ○ **C.** For a given age, each increase of 1 acre in land area is estimated to result in an increase in appraised value by b_1 dollars. For a given land area, each increase in one year in age is estimated to result in a decrease in appraised value by b_2 dollars.

3-2 (cont.)

467.4 thousand dollars

Appraised Value	Property Size	Age
467.4	0.2228	46
365.5	0.2125	58
427.7	0.1604	27
540.3	0.4603	18
401.9	0.2541	40
372.8	0.2201	90
316.9	0.1849	41
747.8	0.5068	2
212.1	0.2282	57
633.7	0.1391	11
348.6	0.1711	55
352.1	0.4214	48
351.7	0.2553	41
275.2	0.1139	12
301.8	0.1683	63
281.2	0.1725	57
396.6	0.3813	43
616.2	0.6582	45
314.4	0.1794	56
360.8	0.1491	74

3-2 solution



a.

Or the LINEST Function in Excel

$$\widehat{Y}_1 = 406.3816 + 460.1572X_1 - 2.7864X_2$$

(impractical to calculate by hand)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.74862856
R Square	0.56044473
Adjusted R Square	0.50873234
Standard Error	94.6639129
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	194239.3065	97119.65325	10.8377273	0.000923878
Residual	17	152341.359	8961.256411		
Total	19	346580.6655			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	406.381589	71.33015986	5.697191614	2.6199E-05	255.8881061	556.875071
Property Size	460.157218	154.5110104	2.978151633	0.008438527	134.1674812	786.146954
Age	-2.7864487	1.026890331	-2.713482306	0.014753669	-4.952997962	-0.6198995

RESIDUAL OUTPUT

Observation	*Predicted Appraised Value*	*Residuals*
1	380.7279745	86.6720255
2	342.5509702	22.9490298
3	404.9566902	22.7433098
4	568.0358785	-27.735879
5	411.8495878	-9.9495878
6	256.8818052	115.918195
7	377.2202596	-60.32026
8	634.016369	113.783631
9	352.5618873	-140.46189
10	439.7385214	193.961479
11	331.8598076	16.7401924
12	466.5423004	-114.4423
13	409.6153278	-57.915328
14	425.3561108	-150.15611
15	308.2797774	-6.4797774
16	326.9311302	-45.73113
17	462.0222397	-65.42224
18	583.8668758	32.3331242
19	332.8926638	-18.492664
20	268.7938227	92.0061773

3-2 solution (cont.)

b.

- A. For a given age, each increase of 1 acre in land area is estimated to result in an increase in appraised value by **1000**b_1 dollars. For a given land area, each increase in one year in age is estimated to result in a decrease in appraised value by **1000**b_2 dollars.

3-2 (cont.)

c. Explain why the regression coefficient, b_0, has no practical meaning in the context of this problem.

- ○ **A.** The interpretation of b_0 has no practical meaning here because it would represent the estimated land area of a new house with no appraised value.
- ○ **B.** The interpretation of b_0 has no practical meaning here because it would represent the estimated appraised value of a new house that has no land area.
- ○ **C.** The interpretation of b_0 has no practical meaning here because it would represent the estimated age of a house with no land area and no appraised value.

d. Predict the appraised value for a house that has a land area of 0.25 acres and is 50 years old.

$ _____ thousand
(Round to two decimal places as needed.)

e. Construct a 95% confidence interval estimate for the the mean appraised value for houses that have a land area of 0.25 acres and is 50 years old.

$ _____ thousand ≤ mean appraised value ≤ $ _____ thousand
(Round to one decimal place as needed.)

f. Construct a 95% prediction interval estimate for the the mean appraised value for houses that have a land area of 0.25 acres and is 50 years old.

$ _____ thousand ≤ mean appraised value ≤ $ _____ thousand
(Round to one decimal place as needed.)

3-2 solution (cont.)

c.

- B. The interpretation of b_0 has no practical meaning here because it would represent the estimated appraised value of a new house ($age = 0$) that has no land area ($size = 0$).

d.

$$\widehat{Y_1} = \mathbf{406.3816 + 460.1572X_1 - 2.7864X_2}$$

$\widehat{Y_1} = 406.3816 + 460.1572 \times 0.25 - 2.7865 \times 50 = 382.10\ thousand$ dollars

e/f.

$$Confidence\ Interval\ (CI) = \hat{y}_0 \pm (t_{\alpha/2} \times S_{\hat{y}_0})$$

$$Perdiction\ Interval (PI) = \hat{y}_0 \pm (t_{\alpha/2} \times \sqrt{S_{(\hat{y}_0)^2} + MSE})$$

From PH stat; 335.8 $thousand \leq \mu_{Y|X} \leq 428.4\ thousand$

From PH stat; 177.1 $thousand \leq Y_X \leq 587.1\ thousand$

(impractical to calculate by hand)

3-5

5. The regression model $\hat{Y}_i = 401.26883 + 466.98135X_{1i} - 2.67307X_{2i}$ predicts appraised value (in thousands of dollars) based on land area of the property (in acres), X_{1i}, and age (in years), X_{2i}, of single-family home i. The data from which the model was constructed are provided. Use the regression model and the data to complete (a) through (c) below.

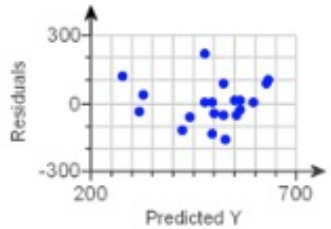
[2] Click the icon to view the table of data.

[3] Click the icon to see the table of critical values of the Durbin-Watson statistic.

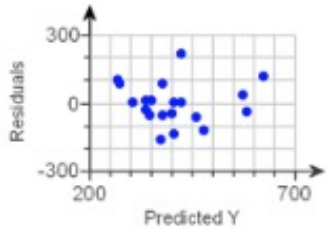
a. Perform a residual analysis on the results.

Plot the residuals versus $\hat{Y}_i$. Choose the correct graph below.

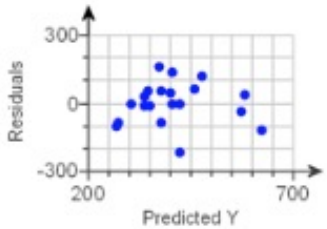
○ **A.**



○ **B.**



○ **C.**



Plot the residuals versus X_{1i}. Choose the correct graph below.

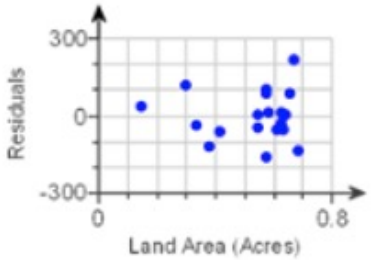
dataset

Home	Appraised Value	Land Area	Age
1	464.5	0.2275	48
2	365.7	0.2158	57
3	425.9	0.1629	20
4	548.5	0.4664	14
5	408.7	0.2547	43
6	372.1	0.2245	89
7	319.1	0.1862	42
8	742.9	0.5041	5
9	215.5	0.2288	51
10	635.4	0.1342	16
11	349.1	0.1732	55
12	355.6	0.4219	46
13	358.7	0.2528	44
14	270.4	0.1185	19
15	306.3	0.1626	65
16	287.4	0.1706	51
17	394.8	0.3825	46
18	613.1	0.6523	49
19	311.8	0.1734	54
20	361.3	0.1445	74

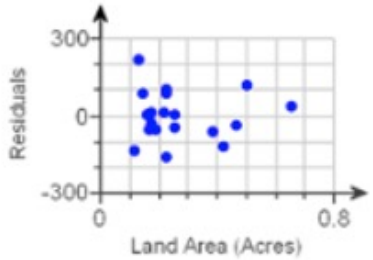
3-5 (cont.)

Plot the residuals versus X_{1i}. Choose the correct graph below.

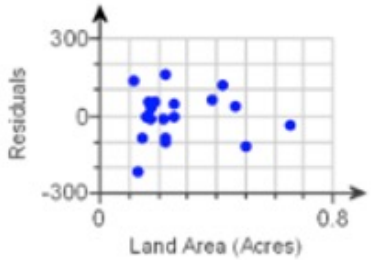
A.



B.

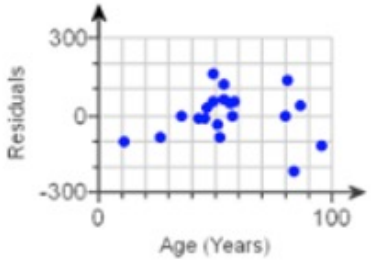


C.

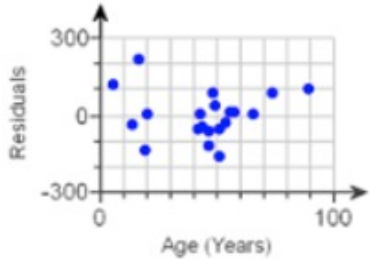


Plot the residuals versus X_{2i}. Choose the correct graph below.

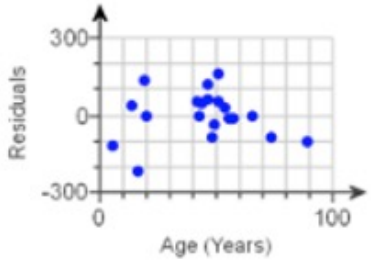
A.



B.



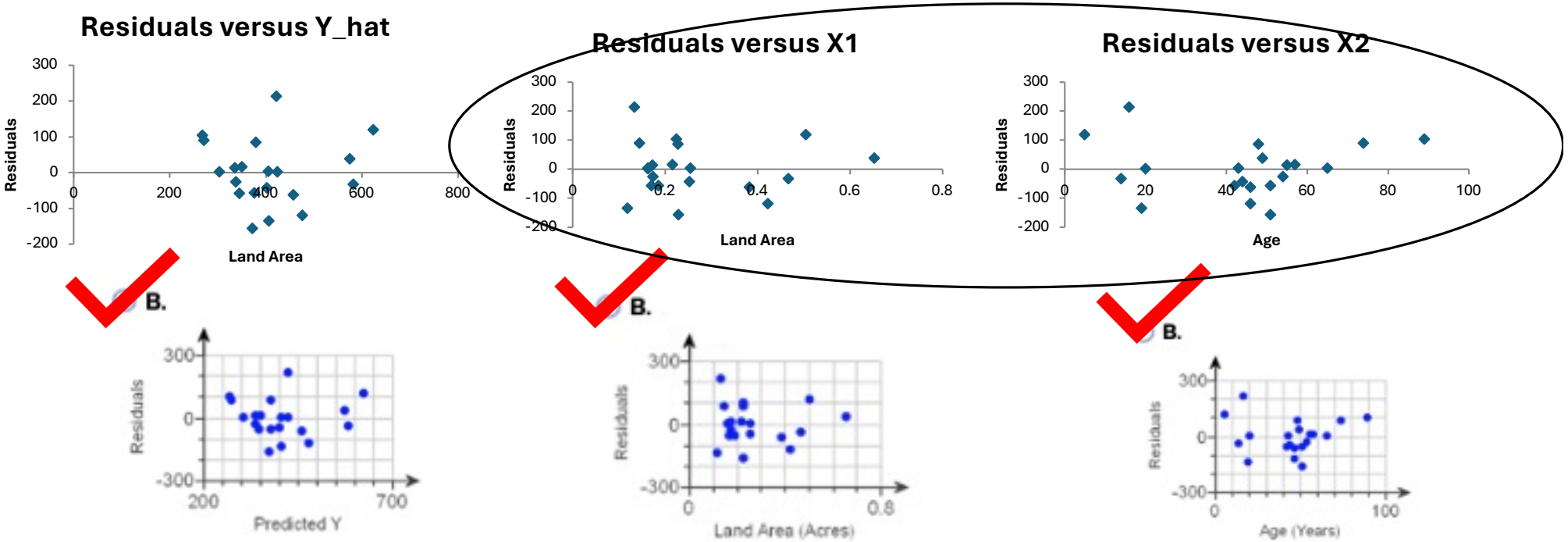
C.



3-5 solution

The residuals are randomly scattered above and below the zero line. No special shape or curved pattern.

This problem requires Excel to solve.



3-5 (cont.)

Make a conclusion regarding the regression model. Choose the correct answer below.

- A. There appears to be a quadratic relationship in each of the residual plots. The model is not adequate.
- B. There is no particular pattern in the plots. The model is not adequate.
- C. There appears to be a quadratic relationship in each of the residual plots. The model appears to be adequate.
- D. There is no particular pattern in the plots. The model appears to be adequate.

c. Are the regression assumptions valid for these data?

- A. Yes, because there is a pattern among the residual plots.
- B. Yes, because there is no particular pattern among the residual plots.
- C. No, because there is a pattern among the residual plots.
- D. No, because there is no particular pattern among the residual plots.

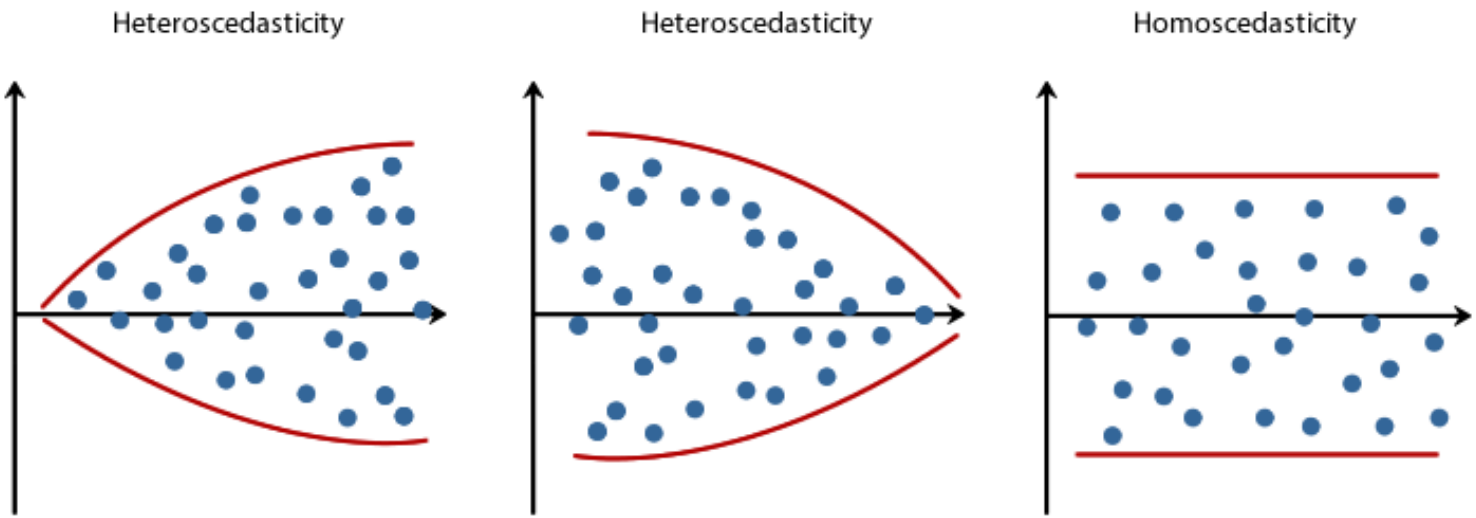
5 Gauss-Markov Assumptions for Simple Linear Model (Wooldridge, p.65)

Linearity

1. *Linear in Parameters* $y = \beta_0 + \beta_1 x_1 + u$
2. *Random Sampling of n observations* $(x_i, y_i) : i = 1, 2, \ldots n$
3. *Sample variation in explanatory variables. x_i's are not all the same value* $x \neq (x_1 = x_2 = x_3 = \ldots = x_n)$

   No multicollinearity
4. *Zero conditional mean.* The error u has an expected value of 0, given any values of the explanatory variable $E(u|x) = 0$
5. *Homoskedasticity.* The error has the same variance given any value of the explanatory variable. $Var(u|x) = \sigma^2$

BLUE
(Best Linear Unbiased Estimator)



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3-5 solution (cont.)

D. The residuals are randomly scattered above and below the zero line. No special shape or curved pattern.

c.

- **B.** Yes, because there is no particular pattern among the residual plots.

3-5 (cont.)

b. If appropriate, perform the Durbin-Watson test, using = 0.05.

Determine the Durbin-Watson statistic. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** D = _____ (Round to two decimal places as needed.)
- **B.** The test is not appropriate.

Determine the critical values. Select the correct choice below and, if necessary, fill in the answer boxes to complete your choice.

- **A.** d_L = _____ and d_U = _____
  (Round to two decimal places as needed.)
- **B.** The test is not appropriate.

Reach a decision. Choose the correct answer below.

- **A.** There is evidence of positive autocorrelation among the residuals.
- **B.** There is no evidence of positive autocorrelation among the residuals.
- **C.** The test is inconclusive.
- **D.** The test is not appropriate.

3-5 solution (cont.)

b.

- The Durbin–Watson test is not appropriate because it is used to test for autocorrelation in residuals that are ordered over time, whereas these data are cross-sectional rather than time-ordered.
- **B/B/D.** The test is not appropriate.

Recap: goodness of fit

Once we have estimated a model, we need to ask two things:

- *How much variation it explains, and whether that explanatory power is statistically meaningful*.
- It evaluates how well a regression model represents your data, typically measured by r^2 to quantify the model's explanatory power (the strength of the fit)
- the **F-test** to verify the overall statistical significance (the reliability of the fit).

Recap: goodness of fit

- **Total Sum of Squares (SST)**: how spread out the actual Y-values are around the mean

$$SST = \sum (Y_i - \bar{Y})^2$$

- **Regression Sum of Squares (SSR)**: how much of the total variation is captured by the model

$$SSR = \sum (\hat{Y}_i - \bar{Y})^2$$

OLS minimizes the SSE

In a perfect world, SSE=0
In reality, SEE as small as possible

- **Error Sum of Squares (SSE)**: $SSE = \sum (Y_i - \hat{Y}_i)^2$

$$\boxed{\mathbf{ST} = \mathbf{SSR} + \mathbf{SSE}}$$

Recap: goodness of fit

- $\boldsymbol{r^2}$ the proportion of total variation in Y explained by the model.

$$\boxed{\mathrm{r}^2 = \frac{SSR}{SST} = \frac{SSR}{SSR + SSE} = 1 - \frac{SSE}{SST}}$$

- A higher r^2 indicates that the model has stronger explanatory power.
- <u>The flaw of plain r^2</u>: adding any new independent variable to a model will **always** make r^2 increase (or stay the same) – even if the variable is useless (e.g., homeowner's shoe size).
- Because of this flaw, you can "cheat" and get a high r^2 just by throwing dozens of irrelevant variables into your regression.

Overfitting!

Recap: goodness of fit

- **Adjusted r^2**: this modified metric introduces a **penalty** for adding unnecessary predictors to your model.

$$r_{adj}^2 = 1 - \left[(1 - r^2)\frac{k}{n - k - 1}\right]$$

- It **only** *increases* if a new variable actually improves the model's predictive power. It will *decrease* if you add statistical "noise".
- Rule: Always use r_{adj}^2 when comparing models that have a different number of independent variables.

Recap: goodness of fit

- The **F-test** tells us whether that explained variation is **statistically significant**. It answers question *"Is the model useful overall?"*
- **Mean Square Regression (MSR)** measures the model's **average** explained variation.

$$MSR = \frac{SSR}{k}$$

where $df_1 = k = number\ of\ independent\ variables/predictors$

- **Mean Square Error (MSE)** measures the model's **average** unexplained variation.

$$MSE = \frac{SSE}{n-k-1}$$

where $df_2 = n - k - 1;\ \mathrm{n} = sample\ size$

Recap: goodness of fit

- The **Degrees of freedom (df)** tell us how many pieces of information are free to vary when we estimate variation.
- For MSR, the degrees of freedom (df_1) are k, because the model use k predictors to explain variation in the dependent variable.
- We begin with *n* data points. Then the regression model estimates *k* slopes and 1 intercept, so a total of *k + 1* parameters are fitted. Once those have been estimated, only *n - k - 1* degrees of freedom are left for the error term.
- So we divide SSE by $n - k - 1$ to get the average unexplained variation left in the residuals.

Recap: goodness of fit

- F-statistics compares explained variation to unexplained variation

$$\boxed{F_{STAT} = \frac{MSR}{MSE} = \frac{SSR/k}{SSE/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)}}$$

- A model may have a higher R^2, but still need the F-test to confirm it is meaningful.
- Hypotheses:

1. Null hypothesis H_0: $\beta_1 = \beta_1 = \cdots = \beta_n = 0$ (none of the predictors have a linear effect on Y)
2. Alternative hypothesis H_1: $at\ least\ one\ \beta_i \neq 0$

Recap: goodness of fit

- Critical threshold from the F-table: $F_{\alpha,k,n-k-1}$
- α is the maximum risk you are willing to take of claiming a result is real (10%, 5%, 1%).
- $df_1 = k; df_2 = n - k - 1$
- This is the cutoff value for the test
- Decision rule:
  - If $F_{STAT} > F_{critical}$, reject H_0
  - Otherwise, do not reject H_0

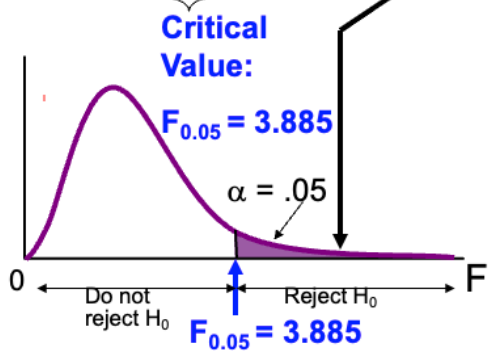
H0: No linear relationship between outcome variable and predictors

H_0: $\beta_1 = \beta_2 = 0$

H_1: β_1 and β_2 not both zero

$\alpha = .05$

$df_1 = 2$ $df_2 = 12$



Test Statistic:

$$F_{STAT} = \frac{MSR}{MSE} = 6.5386$$

Decision:

Since F_{STAT} test statistic is in the rejection region (p-value < .05), reject H_0

Conclusion:

There is evidence that at least one independent variable affects Y

F-table of Critical Values of α = 0.05 for F(df1, df2)

	DF1=1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Recap: goodness of fit

- Limitation: traditional F-tables only provide critical value for predefined significance levels (e.g., $\alpha = 0.05, 0.01$).
- By comparing our test statistic manually, we can only determine a p-value interval (e.g., $we\ only\ know\ p < 0.05, because\ 6.5386 > 3.885$).
- The Solution: Excel's F.DIST.RT function: helps finding the **exact probability** in the upper tail of F-distribution.
- Syntax: $= F.DIST.RT(F_value, \deg_freedom1, \deg_freedom2)$
- Our calculation (p-value): $= F.DIST.RT(6.5386, 2, 12) = 0.012$

The coin toss

1. Set the baseline (The null hypothesis H_0)

- Assumption: Your friend is playing **fair**; it's a normal, unbiased coin.

2. Gather evidence (calculate the probability of pure luck)

- 1 head in a row ($p = 50\%$)
- 3 heads in a row ($p = 0.5 \times 0.5 \times 0.5 = 12.5\%$)

The coin toss

- 5 heads in a row ($p = 0.5 \times 0.5 \times 0.5 \times 0.5 \times 0.5 = 3.13\%$)
- 10 heads in a row ($p = 0.5 \times 0.5 \times \cdots \times 0.5 \times 0.5 = 0.1\%$)

Scenario analysis

- Scenario A: It's a fair coin, and your friend just hit a 1-in-1000 jackpot of luck.
- Scenario B: It's not a fair coin. He is cheating! (*Reject the baseline assumption*)

The coin toss

The final verdict ("confidence")

- If the coin were fair, the probability of observing ten heads in a row would be only about 0.1%.
- Therefore, this result provides ***sufficient evidence*** against the fair-coin hypothesis.

3-3

3. The ANOVA summary table to the right is for a multiple regression model with nine independent variables. Complete parts (a) through (e).

Source	Degrees of Freedom	Sum of Squares
Regression	9	90
Error	27	110
Total	36	200

a. Determine the regression mean square (MSR) and the mean square error (MSE).

MSR = _____ (Round to four decimal places as needed.)

MSE = _____ (Round to four decimal places as needed.)

b. Compute the overall F_{STAT} test statistic.

F_{STAT} = _____ (Round to four decimal places as needed.)

c. Determine whether there is a significant relationship between Y and the two independent variables at the 0.05 level of significance. Choose the correct answer below.

- ○ **A.** There is sufficient evidence of a linear relationship because the test statistic is greater than the critical value.
- ○ **B.** There is insufficient evidence of a linear relationship because the test statistic is less than the critical value.
- ○ **C.** There is insufficient evidence of a linear relationship because the test statistic is greater than the critical value.
- ○ **D.** There is sufficient evidence of a linear relationship because the test statistic is less than the critical value.

3-3 solution

a.

- $MSR = \frac{SSR}{k} = \frac{90}{9} = 10$
- $MSE = \frac{SSE}{n-k-1} = \frac{110}{27} = 4.0741$

b.

- $F_{STAT} = \frac{MSR}{MSE} = \frac{10}{4.0741} = 2.4545$

$From\ the\ F-table$: $F_{(0.05,9,27)}$

c.

- A. There is evidence of a linear relationship because the test statistic is greater than the critical value. (i.e. $2.4545 > \mathbf{2.25}$)

3-3 (cont.)

d. Compute the coefficient of multiple determination, r^2, and interpret its meaning.

r^2 = _____ (Round to four decimal places as needed.)

Interpret the meaning of the coefficient of multiple determination.

The coefficient of multiple determination indicates that _____% of the variation in the (1) _____ variable can be explained by the variation in the (2) _____ variables.

e. Compute the adjusted r^2.

r^2_{adj} = _____ (Round to four decimal places as needed.)

(1) ○ dependent
○ independent

(2) ○ dependent
○ independent

3-3 solution (cont.)

d.

- $r^2 = \frac{SSR}{SST} = \frac{90}{200} = 0.4500$
- The coefficient of multiple determination indicates that **<u>45.00%</u>** of the variation in the **<u>dependent variable</u>** can be explained by the variation in the **<u>independent variables</u>**.

e.

- $r_{adj}^2 = 1 - \left[(1 - r^2)\frac{n-1}{n-k-1}\right] = 1 - (1 - 0.45^2)\frac{36}{27} = 0.2667$

Recap: goodness of fit

- Critical value approach: *"Our calculated F-statistic of 6.5386 exceeds the critical value of 3.885 at the 5% significance level. Consequently, the F-statistic* ***falls into the rejection region****, providing* ***sufficient evidence*** *against the null hypothesis and leading to its rejection."*
- P-value approach: *"Assuming the null hypothesis is true, the probability of observing a result* ***as extreme as, or more extreme than****, our sample data is 0.012. Given this low probability, the data provide* ***sufficient evidence*** *against the null hypothesis, leading to its rejection at the 5% significance level.*
- Simplified verdict: ***"We reject the null hypothesis at the 5% significance level."***

Recap: goodness of fit

- *"Our calculated F-statistic of 2.3400* ***does not*** *exceed the critical value of 3.885 at the 5% significance level. Consequently, the F-statistic* ***does not*** *fall into the rejection region. Therefore, the sample provides* ***insufficient evidence*** *against the null hypothesis, and we* ***do not*** *reject the null hypothesis."*
- *"Assuming the null hypothesis is true, the probability of observing a result* ***as extreme as, or more extreme than****, our sample data is 0.139. Since this probability is greater than 0.05, the data provide* ***insufficient evidence*** *against the null hypothesis. Therefore, we* ***do not reject*** *the null hypothesis at the 5% significance level."*
- Simplified verdict: ***"We <u>do not</u> reject the null hypothesis at the 5% significance level."***

Trap: why don't we say "accept H_0"?

- "Not guilty" does not mean "innocent."
- The defendant may actually be guilty, but the evidence may be insufficient to convict (e.g., the O.J. Simpson case).
- This corresponds to a **Type II error**.
- Likewise, "fail to reject H_0" does not mean "accept H_0."
- H_0 may be false, but the sample may not provide enough evidence against it.

Trap: why don't we say "accept H_0"?

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Justice System - Trial

	Defendant Innocent	Defendant Guilty
Reject Presumption of Innocence (Guilty Verdict)	**Type I Error**	**Correct**
Fail to Reject Presumption of Innocence (Not Guilty Verdict)	**Correct**	**Type II Error**

Trap: why don't we say "accept H_0"?

- Rejecting H_0 does not mean that H_1 has been proven true.
- We may reject H_0 even when H_0 is actually true — this is a **Type I error**.
- Therefore, rejecting H_0 only means that the sample provides strong enough evidence against H_0 at the chosen significance level.

Justice System - Trial

	Defendant Innocent	Defendant Guilty
Reject Presumption of Innocence (Guilty Verdict)	**Type I Error**	**Correct**
Fail to Reject Presumption of Innocence (Not Guilty Verdict)	**Correct**	**Type II Error**

3-4

4. A multiple linear regression model based on a sample of 15 homes is developed to predict appraised property value based on land area of the property and age, in years. The SSR is 57,144 and the SSE is 146,440. Complete parts (a) through (d) below.

a. Determine whether there is a significant relationship between appraised value and the two independent variables (land area and age) at the 0.05 level of significance.

What are the correct hypotheses to test?

H_0: (1) _____

H_1: (2) _____

Calculate the test statistic.

F_{STAT} = _____ (Round to two decimal places as needed.)

Determine the p-value.

The p-value is _____.
(Round to four decimal places as needed.)

State the conclusion.

Since the p-value is (3) _____ than the level of significance, (4) _____ the null hypothesis. There is (5) _____ evidence to conclude there is a linear relationship.

(3) ○ less ○ greater

(4) ○ reject ○ do not reject

(5) ○ sufficient ○ insufficient

3-4 solution

a.

(1) H_0: $\beta_1 = \beta_2 = 0$

(2) H_1: $\beta_j \neq 0 \; for \; at \; least \; one \; of \; j = 1, 2$

$$F_{STAT} = \frac{MSR}{MSE} = \frac{SSR/k}{SSE/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)}$$

where $k = 2, n - k - 1 = 15 - 2 - 1 = 12$

- $F_{STAT} = \frac{57144/2}{146440/12} = 2.34$
- $< F_{(0.05,2,12)} = 3.89 \rightarrow do \; not \; reject \; H_0 \rightarrow$ $no \; linear \; relationship!$

3-4 solution

- p-value? *(impractical to calculate by hand)*
- Excel's <u>F.DIST.RT</u> function
- Syntax: $= F.DIST.RT(F_value, \deg_freedom1, \deg_freedom2)$
- p-value: $= F.DIST.RT(2.34, 2, 12) = 0.139 > 0.05$
- Since the p-value is **<u>greater</u>** than the level of significance, **<u>do not reject</u>** the null hypothesis. There is **<u>insufficient evidence</u>** to conclude there is a linear relationship.

3-4 (cont.)

b. Interpret the meaning of the p-value.

- ○ **A.** The p-value is the probability of getting an overall test statistic equal to or greater than the sample result, assuming there is no linear relationship between the dependent variable and the independent variables.
- ○ **B.** The p-value is the probability that there is a linear relationship between the two independent variables.
- ○ **C.** The p-value is the probability of getting an overall test statistic equal to or less than the sample result, assuming there is no linear relationship between the dependent variable and the independent variables.
- ○ **D.** The p-value is the probability that there is a linear relationship between the dependent variable and at least one of the independent variables.

c. Compute the coefficient of multiple determination, r^2, and interpret its meaning.

r^2 = _____ (Round to four decimal places as needed.)

Interpret the meaning of r^2.

(6)
- ○ land area
- ○ land area and age
- ○ appraised value
- ○ age

(7)
- ○ age.
- ○ land area.
- ○ appraised value.
- ○ land area and age.

The coefficient of multiple determination indicates that _____% of the variation in (6) _____ can be explained by the variation in (7) _____
(Round to two decimal places as needed.)

d. Compute the adjusted r^2.

r^2_{adj} = _____ (Round to four decimal places as needed.)

3-4 solution (cont.)

b.

A. The F-test for overall significance is a **right-tailed test**, finding a "more extreme" result means finding an F-statistics that is exactly the same as or larger than the one you calculated from your sample.

Why are options B and D incorrect?

- Both options describe the p-value as the probability that the hypothesis **itself** is true. However, p-value measures the probability of observing your specific sample data, assuming the null hypothesis is **already true**.

3-4 solution (cont.)

c.

$$r^2 = \frac{SSR}{SST} = \frac{SSR}{SSR + SSE}$$

- $r^2 = \frac{SSR}{SSR+SSE} = \frac{57144}{57144+146440} = 0.2807$
- The coefficient of multiple determination indicates that **<u>28.07%</u>** of the variation in **<u>appraised value</u>** can be explained by the variation in **<u>land area and age</u>**.

d.

$$r_{adj}^2 = 1 - \left[(1 - r^2)\frac{k}{n-k-1}\right] = 1 - (1 - 0.2807)\frac{2}{15-2-1} = 0.1608$$

Recap: t-test

- We just learned the F-test tells us if our overall model is doing a good job. It answers: *"Is the team winning?"*
- But even in a winning team, some players are **MVPs**, and some are just sitting on the bench. The t-test inspects each individual independent variable (X) to see if it's actually contributing to the prediction.

Recap: t-test

How to do the hypothesis test

- H_0: $\beta_i = 0$ (The variable is **useless**, no linear relationship).
- H_1: $\beta_i \neq 0$ (The variable is **useful**, there is a relationship).
- t-statistics

$$t_{STAT} = \frac{b_j - 0}{S_{b_j}}$$

where b_j represents the estimated coefficient (or slope) for the independent variable *j*. And S_{b_j} denotes the standard error of that estimated coefficient. The 0 is the hypothesized value.

Recap: t-test

- **Standard error** (S_{b_j}) measures the uncertainty of the estimated coefficient (b_j). It tells how much the calculated slope might change if we took a completely different sample.
- The formula

$$S_{b_j} = \sqrt{\frac{MSE}{\sum(X_{ij} - \bar{X}_j)^2 \times (1 - r_j^2)}} = \sqrt{\frac{SSE/(n-k-1)}{\sum(X_{ij} - \bar{X}_j)^2 \times (1 - r_j^2)}}$$

- A smaller S_{b_j} means our estimate is precise and reliable.

Not examinable

Recap: t-test

- Computing the t-statistic is not the final step. We still need a **benchmark** to judge it (whether it is big enough to reject the null hypothesis). This is where the **critical value** comes in.
- To use it, we first choose a significance level (usually, $\alpha = 0.05$), and then find the degrees of freedom ($df = n - k - 1$).
- Because t-test is usually a **two-tailed test** for a coefficient, use $\boldsymbol{\alpha/2}$. Look up $t_{(\alpha/2,\boldsymbol{df})}$ in the t-table.

Rule

- If $|t_{STAT}| > t_{(\alpha/2,df)}$, the variable is significant (Reject H_0).
- If $|t_{STAT}| \leq t_{(\alpha/2,df)}$, the variable is not significant (Do not Reject H_0).

Recap: t-test

- Imagine you are the **Football Manager**. You found a rising star and tell the team owner, *"We need to sign this kid, he's game-changer!"*
- The owner doesn't want to waste money. He crosses his arms and says, *"Prove it. I assume he's just lucky, and his actual impact on our team will be ZERO."* (H_0: $\beta_1 = 0$).

Recap: t-test

- To defend your player, you take his observed skill (b_j) and divide it by his inconsistency (S_{b_j}). The higher this number (t_{STAT}), the more his raw cuts through the random noise.

Setup

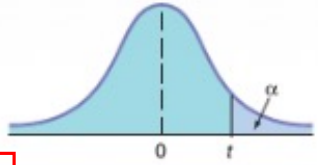
- $b_j = 25\ goals; S_{b_j} = 2.4\ goals; Degrees\ of\ freedom\ (df) = 20\ (= n - k - 1,\ based\ on\ past\ data)$

Recap: t-test

- The boss opens the official league rulebook (the t-table) to set a strict passing score.
- He wants to be 95% sure he isn't getting scammed, using 20 past games as evidence ($df = 20$). He draws a line in the sand at **2.0860**.

Critical Values of t

For a particular number of degrees of freedom, entry represents the critical value of t corresponding to the cumulative probability $(1 - \alpha)$ and a specified upper-tail area (α).



Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	12.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874
26	0.6840	1.3150	1.7056	2.0555	2.4786	2.7787
27	0.6837	1.3137	1.7033	2.0518	2.4727	2.7707
28	0.6834	1.3125	1.7011	2.0484	2.4671	2.7633
29	0.6830	1.3114	1.6991	2.0452	2.4620	2.7564
30	0.6828	1.3104	1.6973	2.0423	2.4573	2.7500
31	0.6825	1.3095	1.6955	2.0395	2.4528	2.7440
32	0.6822	1.3086	1.6939	2.0369	2.4487	2.7385
33	0.6820	1.3077	1.6924	2.0345	2.4448	2.7333
34	0.6818	1.3070	1.6909	2.0322	2.4411	2.7284
35	0.6816	1.3062	1.6896	2.0301	2.4377	2.7238
36	0.6814	1.3055	1.6883	2.0281	2.4345	2.7195
37	0.6812	1.3049	1.6871	2.0262	2.4314	2.7154
38	0.6810	1.3042	1.6860	2.0244	2.4286	2.7116
39	0.6808	1.3036	1.6849	2.0227	2.4258	2.7079
40	0.6807	1.3031	1.6839	2.0211	2.4233	2.7045
41	0.6805	1.3025	1.6829	2.0195	2.4208	2.7012
42	0.6804	1.3020	1.6820	2.0181	2.4185	2.6981
43	0.6802	1.3016	1.6811	2.0167	2.4163	2.6951
44	0.6801	1.3011	1.6802	2.0154	2.4141	2.6923
45	0.6800	1.3006	1.6794	2.0141	2.4121	2.6896
46	0.6799	1.3002	1.6787	2.0129	2.4102	2.6870
47	0.6797	1.2998	1.6779	2.0117	2.4083	2.6846
48	0.6796	1.2994	1.6772	2.0106	2.4066	2.6822
Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005

Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
49	0.6795	1.2991	1.6766	2.0096	2.4049	2.6800
50	0.6794	1.2987	1.6759	2.0086	2.4033	2.6778
51	0.6793	1.2984	1.6753	2.0076	2.4017	2.6757
52	0.6792	1.2980	1.6747	2.0066	2.4002	2.6737
53	0.6791	1.2977	1.6741	2.0057	2.3988	2.6718
54	0.6791	1.2974	1.6736	2.0049	2.3974	2.6700
55	0.6790	1.2971	1.6730	2.0040	2.3961	2.6682
56	0.6789	1.2969	1.6725	2.0032	2.3948	2.6665
57	0.6788	1.2966	1.6720	2.0025	2.3936	2.6649
58	0.6787	1.2963	1.6716	2.0017	2.3924	2.6633
59	0.6787	1.2961	1.6711	2.0010	2.3912	2.6618
60	0.6786	1.2958	1.6706	2.0003	2.3901	2.6603
61	0.6785	1.2956	1.6702	1.9996	2.3890	2.6589
62	0.6785	1.2954	1.6698	1.9990	2.3880	2.6575
63	0.6784	1.2951	1.6694	1.9983	2.3870	2.6561
64	0.6783	1.2949	1.6690	1.9977	2.3860	2.6549
65	0.6783	1.2947	1.6686	1.9971	2.3851	2.6536
66	0.6782	1.2945	1.6683	1.9966	2.3842	2.6524
67	0.6782	1.2943	1.6679	1.9960	2.3833	2.6512
68	0.6781	1.2941	1.6676	1.9955	2.3824	2.6501
69	0.6781	1.2939	1.6672	1.9949	2.3816	2.6490
70	0.6780	1.2938	1.6669	1.9944	2.3808	2.6479
71	0.6780	1.2936	1.6666	1.9939	2.3800	2.6469
72	0.6779	1.2934	1.6663	1.9935	2.3793	2.6459
73	0.6779	1.2933	1.6660	1.9930	2.3785	2.6449
74	0.6778	1.2931	1.6657	1.9925	2.3778	2.6439
75	0.6778	1.2929	1.6654	1.9921	2.3771	2.6430
76	0.6777	1.2928	1.6652	1.9917	2.3764	2.6421
77	0.6777	1.2926	1.6649	1.9913	2.3758	2.6412
78	0.6776	1.2925	1.6646	1.9908	2.3751	2.6403
79	0.6776	1.2924	1.6644	1.9905	2.3745	2.6395
80	0.6776	1.2922	1.6641	1.9901	2.3739	2.6387
81	0.6775	1.2921	1.6639	1.9897	2.3733	2.6379
82	0.6775	1.2920	1.6636	1.9893	2.3727	2.6371
83	0.6775	1.2918	1.6634	1.9890	2.3721	2.6364
84	0.6774	1.2917	1.6632	1.9886	2.3716	2.6356
85	0.6774	1.2916	1.6630	1.9883	2.3710	2.6349
86	0.6774	1.2915	1.6628	1.9879	2.3705	2.6342
87	0.6773	1.2914	1.6626	1.9876	2.3700	2.6335
88	0.6773	1.2912	1.6624	1.9873	2.3695	2.6329
89	0.6773	1.2911	1.6622	1.9870	2.3690	2.6322
90	0.6772	1.2910	1.6620	1.9867	2.3685	2.6316
91	0.6772	1.2909	1.6618	1.9864	2.3680	2.6309
92	0.6772	1.2908	1.6616	1.9861	2.3676	2.6303
93	0.6771	1.2907	1.6614	1.9858	2.3671	2.6297
94	0.6771	1.2906	1.6612	1.9855	2.3667	2.6291
95	0.6771	1.2905	1.6611	1.9853	2.3662	2.6286
96	0.6771	1.2904	1.6609	1.9850	2.3658	2.6280
97	0.6770	1.2903	1.6607	1.9847	2.3654	2.6275
98	0.6770	1.2902	1.6606	1.9845	2.3650	2.6269
99	0.6770	1.2902	1.6604	1.9842	2.3646	2.6264
100	0.6770	1.2901	1.6602	1.9840	2.3642	2.6259
110	0.6767	1.2893	1.6588	1.9818	2.3607	2.6213
120	0.6765	1.2886	1.6577	1.9799	2.3578	2.6174
∞	0.6745	1.2816	1.6449	1.9600	2.3263	2.5758
Degrees of Freedom	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995

Recap: t-test

- If the player's t_{STAT} is higher than 2.0860: He crushes the test.
- Since $|t_{STAT}| = \frac{b_j - 0}{S_{b_j}} = \frac{25-0}{2.4} = |10.42| > 2.08$
- And $p - value \ = 0.000$
- $(Excel\ function's\ syntax: T.DIST.2T(t - value, def_freedom)$
- His skill is statistically real. You sign him $(Reject\ H_0)$!

Recap: t-test

- The owner is finally convinced by the t_{STAT}!
- But before signing the contract, he asks the ultimate question: *'Alright, he's a true MVP. But exactly how many extra goals will he score for us next season?'*"

Recap: t-test

- Rookie scout: *"Boss, the math (our b_j) says he will score 25 goals!"*
- A smart, veteran manager will tell the boss: *"Considering normal performance fluctuations, I am 95% confident he will score between 20 and 30 goals (our CI)."*

Recap: t-test

- The t-test only gives us a “Yes or No” answer. But it doesn’t tell us how much it contributes.
- A **Confidence interval (CI)** gives us a range of realistic values for the true population slope.
- The formula

$$Confidence\ Interval = b_j \pm (t_{\alpha/2} \times S_{b_j})$$

- b_j (our best guess)
- $t_{a/2}$ (critical values from the t-table)
- S_{b_j} (standard error)

Recap: t-test

Setup

- $b_j = 25\ goals; S_{b_j} = 2.4\ goals; Degrees\ of\ freedom\ (df) = 20\ (= n - k - 1,\ based\ on\ past\ data)$
- Goal: create a 95% confidence interval.
- We want 95% confidence → 5% error left over → Split in half = 0.0025 per tail.
- Look at the t-table: $Row\ df = 20, Column\ 0.025$
- $t_{(0.025,20)} = 2.0860 \rightarrow CI = 25 \pm 2.0860 \times 2.4 \rightarrow$ $Lower\ bound: 25 - 5.0064 = 19.99$ $Upper\ Bound: 25 + 5.0064 = 30.01$ $19.99 \leq \beta_1 \leq 30.01$

Z-Distribution (Standard normal)

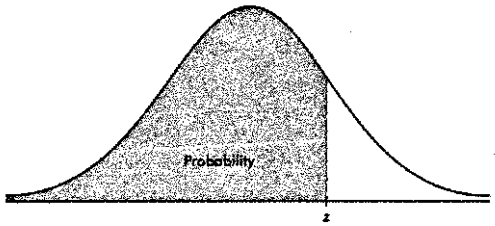
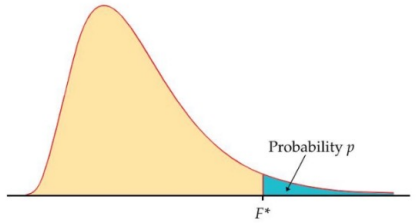


TABLE A: STANDARD NORMAL PROBABILITIES (CONTINUED)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

F-Distribution

or p is the
F^* with
lying to

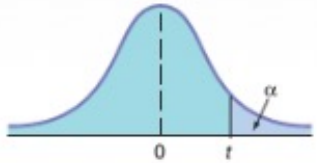


Degrees of freedom in the numerator

2	3	4	5	6	7	8	9
49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86
199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54
799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28
4999.5	5403.4	5624.6	5763.6	5859.0	5928.4	5981.1	6022.5
500000	540379	562500	576405	585937	592873	598144	602284
9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38
19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39
99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39
999.00	999.17	999.25	999.30	999.33	999.36	999.37	999.39
5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24
9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47
30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35
148.50	141.11	137.10	134.58	132.85	131.58	130.62	129.86
4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94
6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90
18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66
61.25	56.18	53.44	51.71	50.53	49.66	49.00	48.47
3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32
5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68
13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16
37.12	33.20	31.09	29.75	28.83	28.16	27.65	27.24
3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96
5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52
10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98
27.00	23.70	21.92	20.80	20.03	19.46	19.03	18.69
3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72
4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82
9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72
21.69	18.77	17.20	16.21	15.52	15.02	14.63	14.33

t-distribution

entry represents
ulative probability



Cumulative Probabilities			
0.95	0.975	0.99	0.995
Upper-Tail Areas			
0.05	0.025	0.01	0.005
6.3138	12.7062	31.8207	63.6574
2.9200	4.3027	6.9646	9.9248
2.3534	3.1824	4.5407	5.8409
2.1318	2.7764	3.7469	4.6041
2.0150	2.5706	3.3649	4.0322
1.9432	2.4469	3.1427	3.7074
1.8946	2.3646	2.9980	3.4995
1.8595	2.3060	2.8965	3.3554
1.8331	2.2622	2.8214	3.2498
1.8125	2.2281	2.7638	3.1693
1.7959	2.2010	2.7181	3.1058
1.7823	2.1788	2.6810	3.0545
1.7709	2.1604	2.6503	3.0123
1.7613	2.1448	2.6245	2.9768
1.7531	2.1315	2.6025	2.9467
1.7459	2.1199	2.5835	2.9208
1.7396	2.1098	2.5669	2.8982
1.7341	2.1009	2.5524	2.8784
1.7291	2.0930	2.5395	2.8609
1.7247	2.0860	2.5280	2.8453
1.7207	2.0796	2.5177	2.8314
1.7171	2.0739	2.5083	2.8188
1.7139	2.0687	2.4999	2.8073
1.7109	2.0639	2.4922	2.7969
1.7081	2.0595	2.4851	2.7874

Comparison of Z, t, and F Distribution

Feature	Z-Distribution	t-Distribution	F-Distribution
Primary Goal	Probability & Description	Inference for Means & Coefficients (Slopes)	Inference for Variances & Overall Models
Core Condition	Population variance (σ^2) is KNOWN.	Population variance (σ^2) is UNKNOWN (estimated by sample s^2).	Comparing the ratio of two independent variances.
Shape of the Curve	Perfect bell shape (symmetric around 0).	Bell-shaped and symmetric, but with "fatter tails" to account for estimation uncertainty.	Right-skewed and non-negative.
Table Logic	Cumulative Probability Table	Critical Value Table	Critical Value Table
Cell Content	Area under the curve(Probability).	x-axis Coordinate (Critical Value).	x-axis Coordinate (Critical Value).
Tail Direction	Usually Left-tail (Cumulative from $-\infty$).	Usually Two-tail (Split $\alpha/2$).	Strictly Right-tail (F is always positive, α).
Degrees of Freedom	None ($df = \infty$).	Yes ($df = n - k - 1$).	Yes (Two: $df_1 = k; df_2 = n - k - 1$).
Typical Statistics	$Z = \frac{X - \bar{X}}{S}$	$t_{STAT} = \frac{b_j - 0}{S_{b_j}}$	$F_{STAT} = \frac{MSR}{MSE}$

Caveat: Alpha vs Alpha/2

- F-test is usually a **one-tailed** (right) test. (1) It <u>cannot be negative</u>; (2) we ***only care if it is large enough***. So the rejection region is all in the right tail. That is why we compare it with alpha.
- A t-statistic <u>can be positive or negative</u>. If you ask whether a coefficient is simply ***different from zero***, you care about both sides.
- So you split the significance level into two tails. That's why a **two-tailed** t-test uses alpha/2 in each tail.

Caveat: one df vs. two df

- A t-test has **one degrees-of-freedom** parameter because it is based on one standardized estimate, while an F-test has **two** because it is <u>a ratio of two variance estimates</u>, one for the numerator and one for the denominator.

$$t_{STAT} = \frac{b_j - 0}{S_{b_j}}$$

a t distribution uses only one df parameter

$$S_{b_j} = \sqrt{\frac{MSE}{\sum(X_{ij} - \bar{X}_j)^2 \times (1 - r_j^2)}} = \sqrt{\frac{SSE/(n-k-1)}{\sum(X_{ij} - \bar{X}_j)^2 \times (1 - r_j^2)}}$$

$$F_{STAT} = \frac{MSR}{MSE} = \frac{SSR/k}{SSE/(n-k-1)}$$

while the f distribution uses two

3-6

6. A regression analysis is performed to predict the durability of a brand of shoe, based on the forefoot shock-absorbing capability (FOREIMP) and the change in impact properties over time (MIDSOLE) for a sample of 20 pairs of shoes. Use the results to complete parts (a) and (b).

Variable	Coefficient	Standard Error	t Statistic	p-Value
INTERCEPT	0.00926	0.06354	0.15	0.8858
FOREIMP	0.04286	0.03376	1.27	0.2213
MIDSOLE	0.21966	0.05958	3.69	0.0018

Click here for page 1 of critical values of t.[4] Click here for page 2 of critical values of t.[5]

a. Construct a 95% confidence interval estimate of the population slope between durability and forefoot shock absorbing capability.

_____ $\leq$ $_1$ $\leq$ _____

(Round to five decimal places as needed.)

b. At the 0.05 level of significance, determine whether each independent variable makes a significant contribution to the regression model. On the basis of these results, indicate the independent variables to include in this model.

Choose the correct hypotheses for FOREIMP below.

- **A.** H_0: $_1 = 0$
  H_1: $_1 \neq 0$
- **B.** H_0: $_1 \neq 1$
  H_1: $_1 = 1$
- **C.** H_0: $_1 \neq 0$
  H_1: $_1 = 0$
- **D.** H_0: $_1 = 1$
  H_1: $_1 \neq 1$

Find the test statistic.

_____ (Round to two decimal places as needed.)

Find the p-value.

_____ (Round to four decimal places as needed.)

3-6 solution

a.

- $df = n - k - 1 = 20 - 2 - 1 = 17$
- $t_{(0.025,17)} = 2.1098$

$$Confidence\ Interval = b_j \pm (t_{\alpha/2} \times S_{b_j})$$

- Thus, 95% confidence interval is $0.04286 \pm (2.1098 \times 0.03376)$
- $-0.02837 \leq \beta_1 \leq 0.11409$

b.

- A. $H_0\colon\ \beta_1 = 0; H_1\colon\ \beta_1 \neq 0$

3-6 solution (cont.)

$$t_{STAT} = \frac{b_j - 0}{S_{b_j}}$$

- Both t_{STAT} and p-value are given in the prompt.
- $|t_{STAT}| = |\frac{0.04286}{0.03376}| = |1.27| < t_{(0.025,17)} = 2.1098$
- p-value? *(impractical to calculate by hand)*
- Excel's <u>T.DIST.2T</u> function
- Syntax: $= T.DIST.2T(t_value, \deg_freedom)$
- $p - value = T.DIST.RT(1.27, 17) = 0.22 > 0.05$

3-6 (cont.)

Is there evidence that the variable FOREIMP contributes to a model already containing MIDSOLE?

- Yes, since the null hypothesis is rejected.
- No, since the null hypothesis is not rejected.
- No, since the null hypothesis is rejected.
- Yes, since the null hypothesis is not rejected.

Choose the correct hypotheses for MIDSOLE below.

- **A.** H_0: $_2 \neq 1$
  H_1: $_2 = 1$
- **B.** H_0: $_2 \neq 0$
  H_1: $_2 = 0$
- **C.** H_0: $_2 = 1$
  H_1: $_2 \neq 1$
- **D.** H_0: $_2 = 0$
  H_1: $_2 \neq 0$

Find the test statistic.

_____ (Round to two decimal places as needed.)

Find the p-value.

_____ (Round to four decimal places as needed.)

3-6 solution (cont.)

- **B.** No, since the null hypothesis is not rejected.
- **D.** $H_0\colon\ \beta_2 = 0; H_1\colon\ \beta_2 \neq 0$
- $|t_{STAT}| = |\frac{0.21966}{0.05958}| = |3.69| > t_{(0.025,17)} = 2.1098$
- $p - value = 0.0018$

3-6 (cont.)

Is there evidence that the variable MIDSOLE contributes to a model already containing FOREIMP?

- ○ Yes, since the null hypothesis is not rejected.
- ○ No, since the null hypothesis is not rejected.
- ○ No, since the null hypothesis is rejected.
- ○ Yes, since the null hypothesis is rejected.

On the basis of these results, indicate the independent variables to include in this model. Choose the correct answer below.

- ○ **A.** Include only the variable MIDSOLE.
- ○ **B.** Include only the variable FOREIMP.
- ○ **C.** Include both variables FOREIMP and MIDSOLE.

3-6 solution (cont.)

- **D.** Yes, since the null hypothesis is rejected.
- **A.** Include variable only the MIDSOLE.

3-7

7. A regression analysis is performed using data for 22 single-family homes to predict appraised value (in thousands of dollars) based on land area of the property (in acres), X_{1i}, and age (in years), X_{2i}, in month i. Use the results below to complete parts (a) and (b) below.

Variable	Coefficient	Standard Error	t Statistic	p-value
Intercept	402.85405	46.64751	8.64	0.0000
Area, X_1	450.03816	118.94585	3.78	0.0013
Age, X_2	− 2.60415	0.66732	− 3.90	0.0010

Click here to view page 1 of the critical values.[6] Click here to view page 2 of the critical values.[7]

a. Construct a 95% confidence interval estimate of the population slope between appraised value and land area.

_____ $\leq {}_1 \leq$ _____

(Round to five decimal places as needed.)

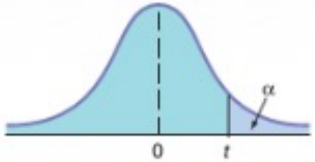
b. At the 0.05 level of significance, determine whether each independent variable makes a significant contribution to the regression model. On the basis of these results, indicate the independent variables to include in this model.

Choose the correct hypotheses for X_1 below.

- **A.** H_0: ${}_1 \neq 0$ H_1: ${}_1 = 0$
- **B.** H_0: ${}_1 = 1$ H_1: ${}_1 \neq 1$
- **C.** H_0: ${}_1 = 0$ H_1: ${}_1 \neq 0$
- **D.** H_0: ${}_1 \neq 1$ H_1: ${}_1 = 1$

Critical Values of t

For a particular number of degrees of freedom, entry represents the critical value of t corresponding to the cumulative probability $(1 - \alpha)$ and a specified upper-tail area (α).



Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005
1	1.0000	3.0777	6.3138	12.7062	31.8207	63.6574
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7531	2.1315	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5177	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874
26	0.6840	1.3150	1.7056	2.0555	2.4786	2.7787
27	0.6837	1.3137	1.7033	2.0518	2.4727	2.7707
28	0.6834	1.3125	1.7011	2.0484	2.4671	2.7633
29	0.6830	1.3114	1.6991	2.0452	2.4620	2.7564
30	0.6828	1.3104	1.6973	2.0423	2.4573	2.7500
31	0.6825	1.3095	1.6955	2.0395	2.4528	2.7440
32	0.6822	1.3086	1.6939	2.0369	2.4487	2.7385
33	0.6820	1.3077	1.6924	2.0345	2.4448	2.7333
34	0.6818	1.3070	1.6909	2.0322	2.4411	2.7284
35	0.6816	1.3062	1.6896	2.0301	2.4377	2.7238
36	0.6814	1.3055	1.6883	2.0281	2.4345	2.7195
37	0.6812	1.3049	1.6871	2.0262	2.4314	2.7154
38	0.6810	1.3042	1.6860	2.0244	2.4286	2.7116
39	0.6808	1.3036	1.6849	2.0227	2.4258	2.7079
40	0.6807	1.3031	1.6839	2.0211	2.4233	2.7045
41	0.6805	1.3025	1.6829	2.0195	2.4208	2.7012
42	0.6804	1.3020	1.6820	2.0181	2.4185	2.6981
43	0.6802	1.3016	1.6811	2.0167	2.4163	2.6951
44	0.6801	1.3011	1.6802	2.0154	2.4141	2.6923
45	0.6800	1.3006	1.6794	2.0141	2.4121	2.6896
46	0.6799	1.3002	1.6787	2.0129	2.4102	2.6870
47	0.6797	1.2998	1.6779	2.0117	2.4083	2.6846
48	0.6796	1.2994	1.6772	2.0106	2.4066	2.6822
Degrees of Freedom	Cumulative Probabilities					
	0.75	0.90	0.95	0.975	0.99	0.995
	Upper-Tail Areas					
	0.25	0.10	0.05	0.025	0.01	0.005

3-7 solution

$$Confidence\ Interval = b_j \pm (t_{\alpha/2} \times S_{b_j})$$

- $df = n - k - 1 = 22 - 2 - 1 = 19$
- A critical value from t-table: $t_{(0.025,19)} = 2.0930$
- $CI = 450.03816 \pm (2.0930 \times 118.94585)$
- $201.08450 \leq \beta_1 \leq 698.99182$

b.

C. $H_0\colon\ \beta_1 = 0; H_1\colon\ \beta_1 \neq 0$

3-7 (cont.)

Find the test statistic.

t_{STAT} = _____
(Round to two decimal places as needed.)

Find the critical value(s).

(1) _____ = _____
(Use a comma to separate answers as needed. Round to four decimal places as needed.)

Reach a conclusion. Choose the correct answer below.

(2) _____ H_0. There (3) _____ sufficient evidence that X_1 makes a significant contribution to the regression model.

(1)
- ○ $\pm t_{/2}$
- ○ $-t$
- ○ t

(2)
- ○ Reject
- ○ Do not reject

(3)
- ○ is not
- ○ is

3-7 solution (cont.)

$$t_{STAT} = \frac{b_j - 0}{S_{b_j}}$$

- $|t_{STAT}| = \left|\frac{450.03816}{118.94585}\right| = |3.78| > t_{(0.025,19)} = 2.0930$
- $\pm t_{(0.025,19)} = \pm 2.0930$
- **<u>Reject</u>** H_0. There **is** sufficient evidence that X_1 makes a significant contribution to the regression model.

3-7 (cont.)

Choose the correct hypotheses for X_2 below.

Find the test statistic.

t_{STAT} = _____
(Round to two decimal places as needed.)

(4) ○ Reject
○ Do not reject

(5) ○ is not
○ is

Reach a conclusion. Choose the correct answer below.

(4) _____ H_0. There (5) _____ sufficient evidence that X_2 makes a significant contribution to the regression model.

On the basis of these results, indicate the independent variables to include in this model. Choose the correct answer below.

○ **A.** Include both variables X_1 and X_2.
○ **B.** Include only the variable X_1.
○ **C.** Include only the variable X_2.

3-7 solution (cont.)

$$t_{STAT} = \frac{b_j - 0}{S_{b_j}}$$

- $|t_{STAT}| = \left|\frac{-2.60415}{0.66732}\right| = |-3.90| = 3.90 > t_{(0.025,19)} = 2.0930$
- **<u>Reject</u>** H_0. There **is** sufficient evidence that X_1 makes a significant contribution to the regression model.
- **A.** Include both variables X_1 and X_2.